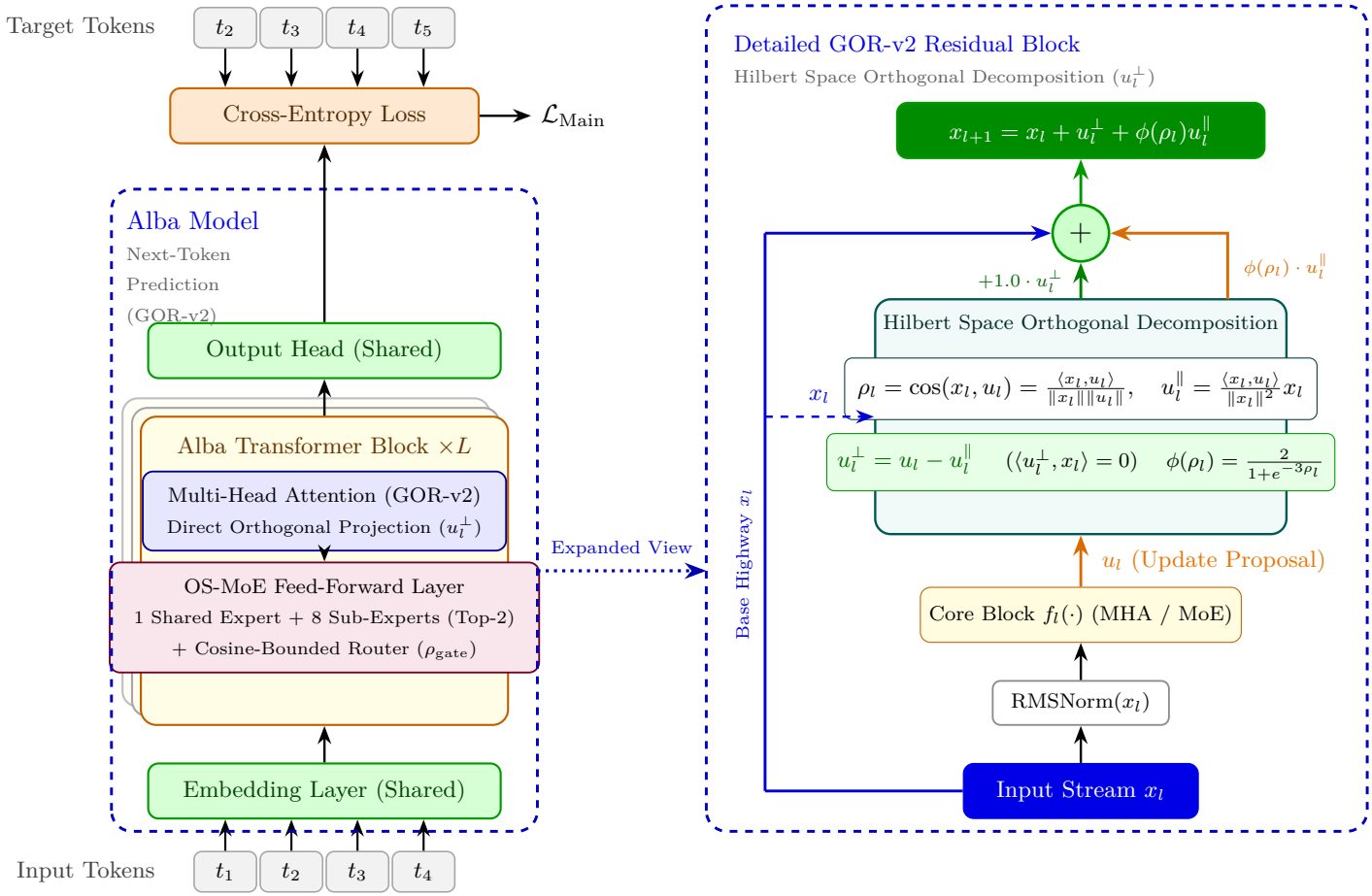
Target Tokens
t_2 t_3 t_4 t_5
Cross-Entropy Loss
$\mathcal{L}_{\text{Main}}$
Alba Model
Next-Token Prediction (GOR-v2)
Output Head (Shared)
Alba Transformer Block $\times L$
Multi-Head Attention (GOR-v2)
Direct Orthogonal Projection ($u_l^{\perp}$)
OS-MoE Feed-Forward Layer
1 Shared Expert + 8 Sub-Experts (Top-2)
+ Cosine-Bounded Router (ρ_{gate})
Embedding Layer (Shared)
Input Tokens
t_1 t_2 t_3 t_4
Expanded View
Detailed GOR-v2 Residual Block
Hilbert Space Orthogonal Decomposition ($u_l^{\perp}$)
$x_{l+1} = x_l + u_l^{\perp} + \phi(\rho_l) u_l^{\parallel}$
$+1.0 \cdot u_l^{\perp}$
$\phi(\rho_l) \cdot u_l^{\parallel}$
Hilbert Space Orthogonal Decomposition
x_l
$\rho_l = \cos(x_l, u_l) = \frac{\langle x_l, u_l \rangle}{\|x_l\| \|u_l\|}, \quad u_l^{\parallel} = \frac{\langle x_l, u_l \rangle}{\|x_l\|^2} x_l$
$u_l^{\perp} = u_l - u_l^{\parallel} \quad (\langle u_l^{\perp}, x_l \rangle = 0) \quad \phi(\rho_l) = \frac{2}{1+e^{-3\rho_l}}$
Base Highway x_l
u_l (Update Proposal)
Core Block $f_l(\cdot)$ (MHA / MoE)
RMSNorm(x_l)
Input Stream x_l